

TSMC Global Facility Network Design Executive Summary

Overview

This project applies the facility network design model discussed in class to TSMC's global semiconductor manufacturing network. The objective is to determine which fabs should be opened and how demand should be assigned in order to minimize total network cost, including facility investment cost, production cost, transportation cost, and unmet-demand penalty.

The model focuses on three advanced-node product groups: Product A (2/3nm), Product B (5nm), and Product C (7nm). These three nodes represent approximately 74% of TSMC's wafer revenue mix based on its 1Q26 technology mix. The demand regions are Silicon Valley, Austin, Shenzhen, Seoul, Munich, and Tokyo. Candidate production locations include Hsinchu, Tainan, Kaohsiung, Arizona, Taichung, and Japan.

Part I. Facility Network Design

Data Estimation Method

The main input parameters were estimated from public information and then converted into the Excel model.

Input	Estimation Method	Model Use
Capacity	TSMC reports approximately 17 million 12-inch equivalent wafers per year. Candidate fab capacities were assigned based on relative manufacturing scale.	Facility capacity constraints
Product Mix	3nm, 5nm, and 7nm account for approximately 74% of TSMC's revenue mix. Applying this ratio to TSMC's estimated annual capacity of 17 million wafers yields 12.58 million wafers of modeled demand. A multiplier of $1/0.74 = 1.351$ is used to link the ABC product group to total wafer capacity.	Product demand split
Total Demand	$17 \text{ million wafers} \times 74\% = 12.58 \text{ million wafers}$, converted into 12,580 model units.	Total demand
Demand Allocation	Demand was allocated to six major semiconductor regions: Silicon Valley 10%, Austin 10%, Shenzhen 30%, Seoul 35%, Munich 10%, and Tokyo 5%.	Regional demand nodes
Freight Cost	Air freight rates were multiplied by shipment weight. The model assumes 1,000 wafers = 320 kg.	Transportation cost matrix
Product Cost	Product A = \$20M, Product B = \$16M, Product C = \$9M per 1,000 wafers.	Variable production cost
Currency	The model reports costs in USD. Currency risk is discussed as a future extension because TSMC operates across multiple currency regions.	Financial interpretation

Freight costs vary by route. For example, shipping from Hsinchu to Shenzhen costs \$793.6 per 1,000 wafers, while Hsinchu to Austin costs \$2,406.4 per 1,000 wafers. These values were added to production costs to form the variable cost matrix in the Excel model.

Network Design Results

The optimization model selected the following facilities:

Facility	Decision
Hsinchu	Open
Tainan	Open
Taichung	Open
Kaohsiung	Closed
Arizona	Closed
Japan	Closed

The selected network concentrates production in Taiwan because Hsinchu, Tainan, and Taichung provide the lowest combination of fixed cost, production cost, and feasible technology capability.

Capacity and Cost Results

Metric	Value
Total Demand	12,580
Opened Capacity	5,400
Total Available Capacity	7,920
Unmet Demand	7,180
Fixed Facility Cost	\$36.0B
Production & Transportation Cost	\$70.9B
Unmet Demand Penalty	\$215.4B
Total Cost	\$322.3B

The results show that available capacity is not sufficient to satisfy all projected demand. Approximately 57% of demand remains unmet. The unmet-demand penalty accounts for about 67% of total cost, making capacity shortage the dominant issue in the network.

From a managerial perspective, this penalty can be interpreted as the financial impact of lost sales opportunities, delayed customer deliveries, and potential market-share losses to competing foundries when demand cannot be fulfilled.

Managerial Insights

First, Arizona was not selected even though it is closer to North American demand centers. The reason is that Arizona's fixed cost is much higher than Taiwan-based facilities. Its transportation savings are not large enough to offset the additional investment cost. This shows that in semiconductor manufacturing, facility cost can dominate transportation cost. We also display Japan as we believe it is a good alternative for the future as demand is unmet even opening the other fabs.

Second, capacity is the main bottleneck. Production and transportation costs are approximately \$70.9B, while the unmet-demand penalty is \$215.4B. This means that expanding

advanced-node capacity would likely create more value than small improvements in transportation efficiency.

Third, Product C demand is fully satisfied, while Product A experiences the largest shortage. This result reflects the model's tendency to allocate limited capacity toward products with lower production costs and feasible manufacturing locations.

The shortage penalty was calibrated through model testing and sensitivity analysis. A value of \$30M per 1,000 wafers was selected because it is higher than the production cost of all product categories and encourages demand fulfillment when economically reasonable. Sensitivity testing also showed that using \$50M did not change the facility-opening decision, which suggests that the selected network is relatively robust to the penalty assumption.

Part II. Limitations and Future Extensions

Although the model provides useful insights, several real-world issues are not fully captured.

Issue Not Captured	Why It Matters	Possible Model Extension
Capacity Expansion	Fab capacity cannot be expanded instantly, but long-term investment decisions are critical.	Add multi-period capacity expansion decisions, investment budgets, and construction lead times.
Demand Uncertainty	Semiconductor demand changes with AI, smartphones, HPC, and automotive cycles.	Add demand scenarios such as base, high-growth, and low-growth cases.
Technology Migration	Fabs gradually move from older nodes to advanced nodes.	Add technology upgrade decisions, migration costs, and yield ramp-up effects.
Geopolitical Risk	Taiwan concentration, export controls, and trade restrictions affect supply resilience.	Add regional risk costs, disruption scenarios, or diversification constraints.
Currency Risk	TSMC operates globally and incurs costs in multiple currencies such as USD, TWD, JPY, and EUR. Exchange-rate fluctuations can affect production costs, transportation expenses, and profitability.	Incorporate exchange-rate scenarios, currency-adjusted cost parameters, or financial hedging decisions into the network design model.
Customer Priority	Not all demand has the same strategic value.	Add customer-specific service levels and differentiated shortage penalties.
ESG Constraints	Semiconductor fabs require large amounts of electricity, water, and other resources.	Add carbon, renewable energy, water, and environmental compliance constraints.

Conclusion

The model identifies Hsinchu, Tainan, and Taichung as the preferred facility network under the current assumptions. While transportation costs affect assignment decisions, the results clearly show that capacity shortage is the most important driver of total cost. Therefore, TSMC's future network strategy should focus not only on minimizing transportation cost, but also on expanding advanced-node capacity, managing demand uncertainty, managing currency exposure, and improving geographic resilience.

Appendix . Model Formulation and Solver Implementation

Decision Variables

The model includes facility opening decisions, production allocation decisions, and unmet-demand variables.

Objective Function

The Solver minimizes total cost:

Total Cost = Fixed Facility Cost + Production Cost + Transportation Cost + Unmet Demand Penalty

The shortage penalty is included because total demand exceeds available production capacity under the current assumptions. It represents the economic impact of failing to satisfy customer demand, including lost revenue, delayed deliveries, reduced service levels, and the possibility that customers shift orders to competing foundries.

The penalty value was calibrated through sensitivity testing and model experimentation. A value of \$30 million per 1,000 wafers was selected because it is larger than the production cost of all product categories and encourages demand fulfillment whenever economically reasonable, while still allowing unmet demand when capacity constraints become binding.

Major Constraints

Capacity constraints ensure that production assigned to each fab does not exceed available capacity. Demand constraints track whether demand is satisfied or remains unmet. Facility activation constraints ensure that production can only occur when a facility is opened. Technology constraints prevent infeasible product-fab combinations, such as Taichung producing Product A or Arizona producing Product C.

Excel Solver Setup

Solver Element	Excel Setting
Objective Cell	E46
Objective	Minimize total cost
Production Variables	B21:S26
Facility Binary Variables	T21:T26
Capacity Constraints	B31:B36 ≥ 0
Demand Balance	B37:S37 ≥ 0
Facility Decisions	T21:T26 = Binary
Non-negativity	B21:S26 ≥ 0
Technology Restrictions	Infeasible production cells fixed at 0

Final Solver Result

Result	Value
Open Facilities	Hsinchu, Tainan, Taichung
Closed Facilities	Kaohsiung, Arizona, Japan
Total Demand	12,580 (1 Batch=1000 Units)
Opened Capacity	5,400 (1 Batch=1000 Units)
Total Available Capacity	7,920 (1 Batch=1000 Units)
Unmet Demand	7,180 (1 Batch=1000 Units)
Penalty Cost	\$215.4B
Total Cost	\$322.3B

Sources

TSMC Fab Capacity: https://www.tsmc.com/english/dedicatedFoundry/manufacturing/fab_capacity

TSMC 1Q26 Investor Presentation:

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Semiconductor Wafer Cost Reference: <https://siliconanalysts.com/guide/semiconductor-costs>

Air Freight Cost Reference: <https://suaidglobal.com/insights/air-freight-cost-per-kg/>